

Article

Synthesis of omega-Oxo Amino Acids and trans-5-Disubstituted Proline Derivatives using Cross-methathesis of Unsaturated Amino Acids

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Synthesis of ω -Oxo Amino Acids and *trans*-5-Disubstituted Proline Derivatives using Cross-methathesis of Unsaturated Amino Acids

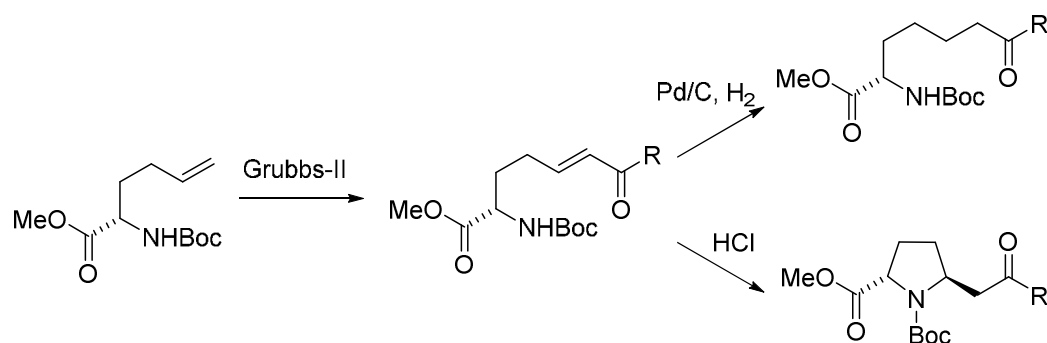
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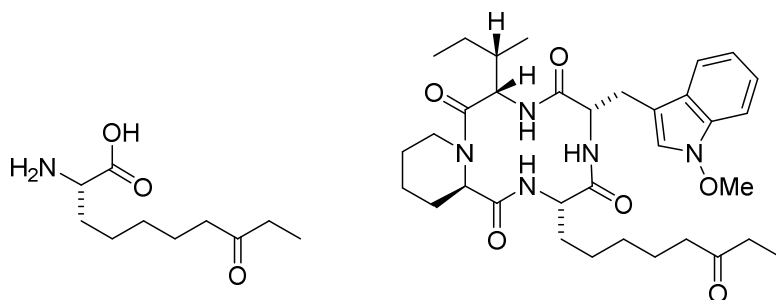
Abstract



A range of 7-oxo, 8-oxo and 9-oxo amino acids, analogues of 8-oxo-2-aminodecanoic acid, one of the key components of the cyclic tetrapeptide apicidin, has been prepared by a three-step process involving copper-catalysed allylation of serine-, aspartic acid- and glutamic acid-derived organozinc reagents, followed by cross-metathesis of the resulting terminal alkenes with unsaturated ketones, and hydrogenation. The intermediate 7-oxo-5-enones underwent a highly diastereoselective ($dr \geq 96:4$) acid-catalyzed aza-Michael reaction to give *trans*-2,5-disubstituted pyrrolidines, 5-substituted proline derivatives. The aza-Michael reaction was first observed when the starting enones were allowed to stand in solution in deuteriochloroform, but can be efficiently promoted by catalytic amounts of dry HCl.

Introduction

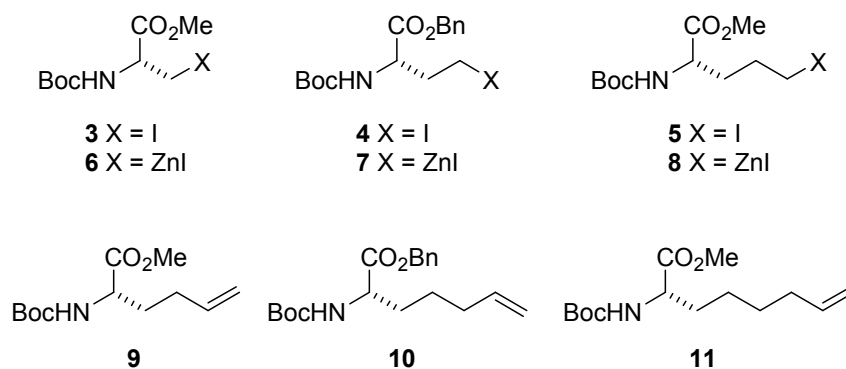
The synthesis of 8-oxo-2-aminodecanoic acid (Aoda) **1**, one of the key components of the cyclic tetrapeptide apicidin **2**,¹⁻³ has attracted considerable attention, as a result of the biological activity of apicidin,^{4,5} and closely related analogues.⁶⁻¹⁰ A number of approaches to the synthesis of Aoda **1** have been adopted, including the use of chiral pool starting materials,^{9,11-14} and the use of chiral auxiliary groups.¹⁵⁻¹⁸ Of most direct relevance to the topic of this paper is the report on the use of radical addition of chiral non-racemic amino acid fragments to enones.¹² However, there is as yet no general approach to a range of simple analogues of 8-oxo-2-aminodecanoic acid in which the position of the ketone and the length of the side-chain can be straightforwardly varied.



2-amino-8-oxo-decanoic acid (Aoda), **1**

Apicidin, **2**

We have developed a direct approach to the synthesis of enantiomerically pure α -amino acids using a family of organozinc reagents **6-8**, each prepared from the corresponding alkyl iodide **3-5**.¹⁹ Zinc reagents **6-8** can undergo Negishi cross-coupling with a variety of coupling partners.¹⁹ Copper-promoted reaction of zinc reagent **6** with a range of electrophiles, including allylic halides, is possible.²⁰ Specifically copper-catalyzed reaction of zinc reagent **3** with allyl chloride gave protected butenylglycine **9**,²¹ and the protected pentenylglycine **10** has also been prepared by related chemistry.²² We considered that extension of this allylation reaction to the zinc reagent **8** should allow access to the homologous alkene **11**.

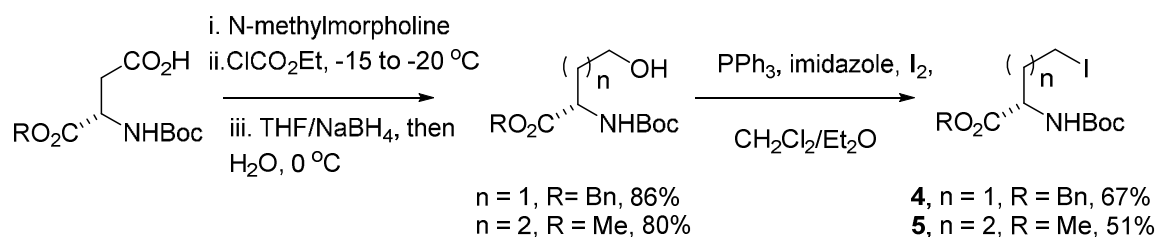


There is significant literature precedent that amino acids incorporating a terminal alkene in the sidechain can undergo cross-metathesis reaction^{23,24} with simple alkenes^{25,26} including with electron-deficient alkenes.^{27,28} A recent paper describes application of this approach to cyclic tetrapeptide derivatives.²⁹ It therefore appeared entirely reasonable that the terminal alkenes **9-11** might undergo cross-metathesis with simple enones. Subsequent hydrogenation would be expected to give the desired analogues of 8-oxo-2-aminodecanoic acid, in a straightforward and flexible manner.

Results and Discussion

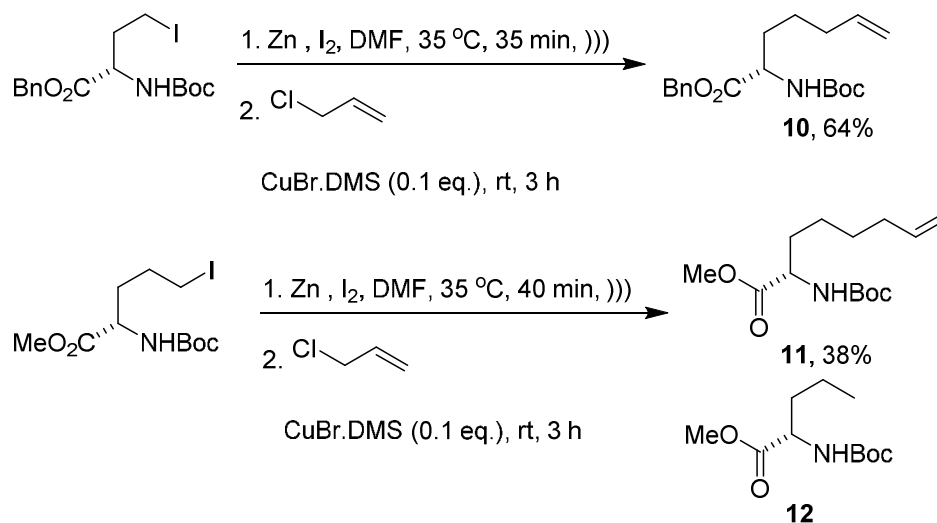
Synthesis of the iodide **3** was carried out according to our previously reported methods from appropriately protected derivatives of serine.³⁰ Iodides **4** and **5** were prepared from protected aspartic and glutamic acids by reduction *via* the mixed anhydride,³¹ an improvement on the method using N-hydroxysuccinimide activation,³² followed by standard conversion of the primary alcohol to the iodide (Scheme 1).

Scheme 1

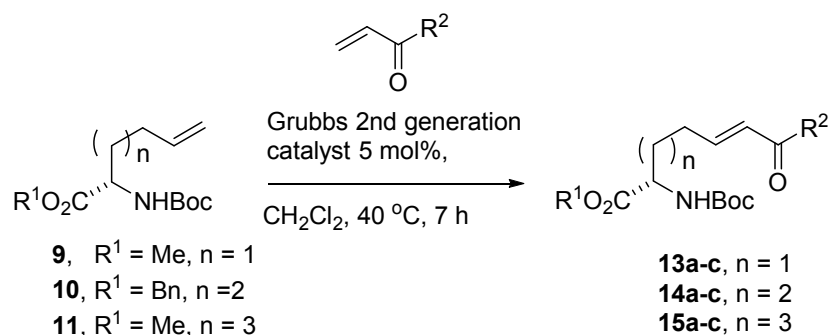


Conversion of the iodide **3** into the corresponding zinc reagent **6** using iodine to activate the zinc metal prior to insertion, and copper catalysed allylation using allyl chloride, gave the expected product **9** in slightly higher yield (75%) than our previously reported method.²¹ Others have successfully prepared **9** using this general approach.³³ Conversion of each of the two iodides **4** and **5** into the corresponding zinc reagents using ultrasonication, followed by allylation using allyl chloride in the presence of CuBr.DMS (0.1 eq.), gave the corresponding allylated products in yields of **10** (64%) and **11** (38%), respectively (Scheme 2). While the yield of **10** is comparable to that obtained for **9**, the yield of **11** was disappointing. In the latter case, the mass balance was the protonated zinc reagent **12**, and attempts to improve this yield, including omitting the use of sonication, were not successful.

Scheme 2



Cross-metathesis reactions of each of the terminal alkenes **9-11** using the Grubbs 2nd generation catalyst with a range of unsaturated ketones gave excellent yields of the expected products **13-15** (Scheme 3, Table 1).

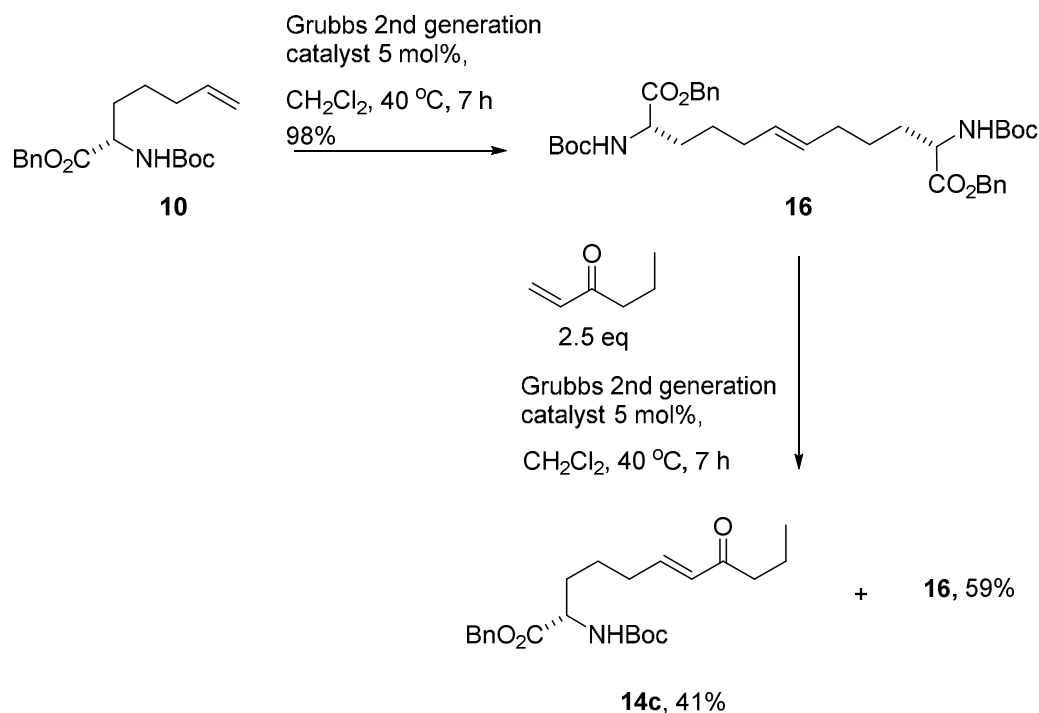
Scheme 3. Cross-metathesis of terminal alkenes**Table 1.** Cross-metathesis of terminal alkenes

Substrate	n	R ¹	R ²	Enone equiv.	Product	Yield %
9	1	Me	CH ₃	3.0	13a	91
9	1	Me	C ₂ H ₅	3.0	13b	92
9	1	Me	C ₃ H ₇	2.5	13c	90
10	2	Bn	CH ₃	3.0	14a	89
10	2	Bn	C ₂ H ₅	5.0	14b	90
10	2	Bn	C ₃ H ₇	2.5	14c	86
11	3	Me	CH ₃	2.5	15a	82
11	3	Me	C ₂ H ₅	2.5	15b	82
11	3	Me	C ₃ H ₇	2.5	15c	91

Homodimers of **9-11** were detected in all the crude reaction mixtures by MS analysis. This observation was not unexpected, since terminal alkenes are known to undergo rapid homodimerization under similar conditions.^{34,35} When the alkene **10** was subjected to Grubbs 2nd generation catalyst in the absence of enone, the homodimer **16** was isolated in excellent yield (98%). Subjection of **16** to the standard cross-metathesis conditions with 1-hexen-3-one gave the expected product **14c** (41%), together with recovered homodimer **16** (59%) (Scheme 4). Since the homodimer was not completely consumed under the reaction

conditions used for the initial cross-metathesis, we can conclude that it is probably not an intermediate in that process even though it is a substrate for the cross-metathesis reaction.

Scheme 4



Hydrogenation of each of the enones **13-15** was carried out under standard conditions, leading to the desired protected amino acids **17-19** (Scheme 5, Table 2). In the case of the three benzyl esters **14a-c**, the final isolated product was the corresponding free carboxylic acid **18a-c**, in principle ready for incorporation into a peptide.

Scheme 5

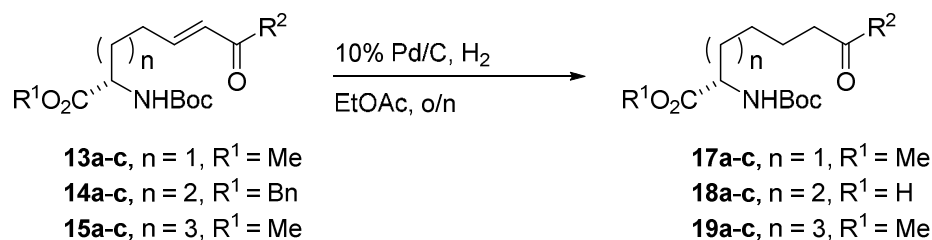


Table 2. Hydrogenation of cross-metathesis products

Substrate	n	R ¹	R ²	Product	R ¹	Yield %
13a	1	Me	CH ₃	17a	Me	94
13b	1	Me	C ₂ H ₅	17b	Me	99
13c	1	Me	C ₃ H ₇	17c	Me	98
14a	2	Bn	CH ₃	18a	H	90
14b	2	Bn	C ₂ H ₅	18b	H	89
14c	2	Bn	C ₃ H ₇	18c	H	98
15a	3	Me	CH ₃	19a	Me	94
15b	3	Me	C ₂ H ₅	19b	Me	99
15c	3	Me	C ₃ H ₇	19c	Me	99

These results demonstrate that it is possible to prepare a range of simple analogues of 8-oxo-2-aminodecanoic acid in which the position of the ketone and the length of the side-chain can be straightforwardly varied, simply by choice of starting amino acid (serine, aspartic acid or glutamic acid), and enone.

During the process of characterization of the enones **13a-c**, and in particular upon allowing a solution of each of these enones to stand in CDCl₃, they were each converted in high yield into the corresponding pyrrolidines **20-22**, in an intramolecular aza-Michael reaction. Use of purified CDCl₃, in which any HCl present in the CDCl₃ was removed by passage through UG1 alumina, prevented the aza-Michael reaction from occurring, allowing characterization of the enones **13a-c**. In separate experiments, each of the enones **13a-c** was separately treated with catalytic amounts of dry HCl in CH₂Cl₂ which resulted in efficient cyclization to give the same pyrrolidines **20-22** already observed (Scheme 6, Table 3). This established

that the aza-Michael reaction was acid catalysed, something that has been observed previously.³⁶⁻³⁸ What was striking about each of the three pyrrolidines **20-22** resulting from acid-catalyzed cyclization is that they appeared to be formed with very high diastereoselectivity, with dr values ranging from 96:4 to 98:2 (determined by GC analysis, and also corroborated by NMR in the case of **20a/20b**). X-ray diffraction analysis of the major product obtained by cyclisation of **13b** provided a definitive answer, showing that the major isomer was of *trans*-configuration **21b**. In order to exclude the possibility that we had inadvertently selected a crystal of the minor isomer, the ¹H NMR of the specific crystal used for X-ray analysis was recorded. Although the concentration of the NMR sample was low, the spectrum matched closely that of the bulk material. The ¹H NMR spectra of the products formed from acid catalysed cyclisation of **13a** and **13c** were essentially identical to the ¹H NMR spectrum of **21b** (apart from the signals due to the different side-chains) which confirms that the products therefore have structures **20b** and **22b**, respectively.

Scheme 6

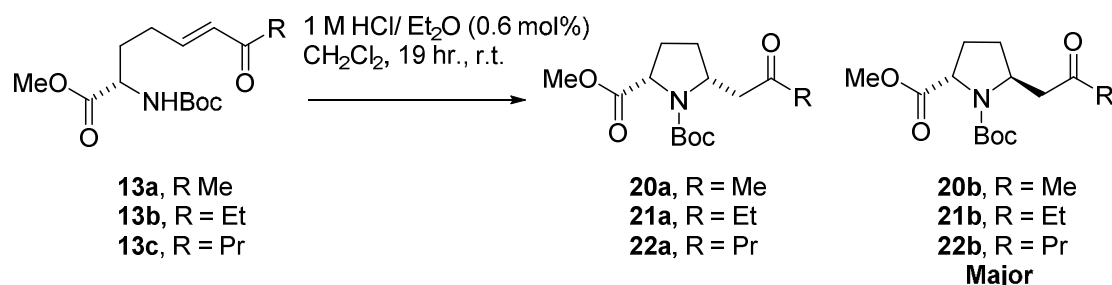


Table 3. Acid-Catalyzed Cyclization of Amino Enones 13

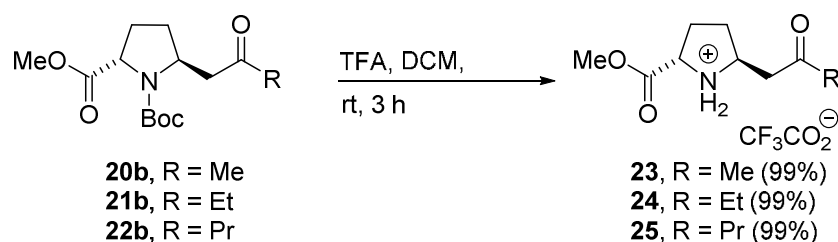
Substrate	R	Product	Conversion	*d.r. (<i>cis</i> : <i>trans</i>)
13a	CH ₃	20a/b	Complete	0.04 : 0.96 ^{a,b}
13b	C ₂ H ₅	21a/b	Complete	0.02 : 0.98 ^b

13c	C ₃ H ₇	22a/b	96 (4% SM)	0.02 : 0.98 ^b
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^a Ratio determined by ¹H NMR; ^b ratio determined with GC.

Boc deprotection of each of the compounds **20b**, **21b**, **22b** using trifluoroacetic acid (Scheme 7) gave the corresponding trifluoroacetate salts **23**, **24**, and **25** in essentially quantitative yield. Each of the trifluoroacetate salts **23**, **24**, and **25** was determined to be of *trans* configuration by X-ray diffraction. Since the structure of **21b** had already been established as the *trans*-pyrrolidine, this demonstrated that the deprotection reaction had proceeded without influencing the stereochemistry (for example by promoting a reversible aza-Michael process), which means that the assignments already made by comparison of the ¹H NMR spectra of **20b** and **22b** with **21b** are confirmed.

Scheme 7



A similar process has been reported previously, in which the cross-metathesis of Z-protected unsaturated amines with enones in the presence of a Lewis acid (BF₃·Et₂O) leads to 2,5-disubstituted pyrrolidines with moderate levels of stereoselectivity (in the range 2:1 to 6:1 in favor of the *trans*-isomer).³⁹ Strong acid catalysis (TfOH) has also been used to promote the cyclization of a related Cbz-protected amino enone, again with a moderate levels of stereoselectivity (39:61 in favor of the *trans*-isomer).³⁸ The high levels of diastereoselectivity in the formation of *trans*-2,5-disubstituted pyrrolidines that we have observed were therefore not expected on the basis of this literature precedent. The two

principal differences between our substrates and those previously reported are the nature of the nitrogen protecting group, and the presence of a carbomethoxy group. In addition, the choice of acid catalyst may be important. Further studies on the influence of these features appear to be warranted.

Experimental Section

HRMS measurements were carried out using Electrospray ionisation, with a TOF mass analyzer. IR spectra were recorded as thin films. Compound **3** was prepared by the literature method.³⁰ The preparation of compound **4** was carried out by the literature method,³¹ but on a substantially larger scale than that reported. In our hands, **4** was isolated as a yellow crystalline solid (mp 55-56 °C), as we had previously reported (mp 54-55 °C),³² rather than as an oil.³¹ All other data for **4** matched that reported.^{31,32} Compound **5** (the methyl ester)⁴⁰ was prepared by the general method reported for the synthesis of the corresponding benzyl ester.³¹ GC analysis of compounds **20** – **22** was carried out using a Phenomenex ZB-5 column (0.25 mm i.d. × 30 m., film thickness = 250 µm.), oven temperature = 145 °C isothermal; carrier gas: H₂ at 1.4 mL/min.; injection: 250 °C/split = 34.7:1; detection: FID at 300 °C.

General Procedure A: Allylation reactions

A two-necked round bottomed flask was charged with a magnetic stirrer bar was fitted with a rubber septum and three-way tap. The flask was flame-dried under vacuum and backfilled with nitrogen three times. Zinc dust (see each procedure for amount) was added, flame-dried and again evacuated and backfilled with nitrogen three times, with continuous stirring. The flask was allowed to cool, dry DMF (1 mL/1 mmol of alkyl iodide) was added via syringe, and the heterogeneous mixture stirred vigorously. Iodine (see each procedure for

amount) was added by rapid removal and replacement of the three-way tap under a stream of nitrogen. The mixture was stirred for 1-2 minutes, until the solution was colorless. The alkyl iodide (1.0 mmol) was added by rapid removal and replacement of the three-way tap under a stream of nitrogen (in the case of compound **5**, the alkyl iodide was dissolved in DMF and added by syringe). The mixture was stirred and an exotherm was observed stirring continued for a further 50 minutes at r.t. or 35-40 minutes at 35 °C with sonication; these details are specified with each example. The solid zinc dust was then allowed to settle, giving a clear supernatant. During the activation period, a separate two-necked round bottomed flask fitted with a magnetic stirrer bar, rubber septum and three-way tap was flame-dried under vacuum and backfilled with nitrogen three times. The flask was allowed to cool, CuBr.DMS (0.1 eq. relative to alkyl iodide) was added and gently heated then evacuated and backfilled with nitrogen until the CuBr.DMS changed appearance from a grey-brown to light green powder. The flask was allowed to cool, before adding dry DMF (0.6 mL/1 mmol of alkyl iodide) and allyl chloride (see each procedure for amount) dropwise *via* syringe. The mixture was stirred at room temperature for about 5 minutes, at which point the supernatant from the solution containing the organozinc reagent was added dropwise *via* syringe, and the reaction mixture was stirred at room temperature for 3 h. The crude reaction mixture was directly applied to SiO₂ column, using a gradient of 20-30% EtOAc in petroleum ether.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)hex-5-enoate (9)

General procedure **A** using zinc dust (1.95 g, 30 mmol, 2.5 eq.), iodine (0.6 g, 2.4 mmol, 0.2 eq.), **3** (3.94 g, 12 mmol, 1 eq.), CuBr.DMS (0.246 g, 1.2 mmol, 0.1 eq.), and allyl chloride (1.36 mL, 16.8 mmol, 1.4 eq.) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)hex-5-

enoate **9** (2.2 g, 9 mmol, 75 %) as a colorless oil. Zinc insertion took 50 minutes at r.t. $[\alpha]_D -17.0$ (c 1.0, MeOH), lit.³³: $[\alpha]_D -20.7$ (c 0.97, MeOH); $R_f = 0.57$ (20% EtOAc in petroleum ether); $^1\text{H NMR}$ (400 MHz, CDCl_3) δ ppm: 5.80 (1 H, ddt, $J = 16.9, 10.3$ and 6.6 Hz), 4.98-5.08 (3 H, m), 4.28-4.37 (1 H, m), 3.74 (3 H, s), 2.06-2.17 (2 H, m), 1.85-1.96 (1 H, m), 1.65-1.77 (1 H, m), 1.44 (9 H, s); $^{13}\text{C NMR}$ (100 MHz, CDCl_3) δ ppm: 173.3, 155.3, 136.9, 115.7, 79.8, 52.9, 52.3, 31.9, 29.5, 28.3; IR (cm^{-1}), 3370, 1742, 1712, 1516, 1451, 1365, 1254, 1168; m/z (ES+) found MH^+ : 244.1549, $\text{C}_{12}\text{H}_{22}\text{NO}_4$ requires 244.1549.

Benzyl (2S)-2-([(tert-butoxy)carbonyl]amino)hept-6-enoate (10)²²

General procedure A using zinc dust (1.17 g, 18 mmol, 3 eq.), iodine (0.335 g, 1.32 mmol, 0.22 eq.), **4** (2.5 g, 6 mmol, 1 eq.), CuBr.DMS (0.123 g, 0.6 mmol, 0.1 eq.), and allyl chloride (0.7 mL, 8.4 mmol, 1.4 eq.). Zinc insertion took 35 minutes with sonication at 35 °C. Purification by column chromatography (20% EtOAc in petroleum ether) followed by preparative HPLC (XBridge Prep OBD C18 5 μm 19 mm id x 250 mm, using 30:70 water/acetonitrile, at a flow rate of 17 mL.min⁻¹ and UV detection at 210 nm) gave benzyl (2S)-2-([(tert-butoxy)carbonyl]amino)hept-6-enoate **10** (1.28 g, 3.8 mmol, 64%) as a colorless oil ($t_R = 8$ -10 min). $[\alpha]_D -4.0$ (c 1, CHCl_3) lit.²²: $[\alpha]_D -5.5$ (c 1.4, CH_2Cl_2), $R_f = 0.5$ (20% EtOAc in petroleum ether). $^1\text{H NMR}$ (400 MHz, CDCl_3) δ ppm: 7.31-7.40 (5 H, m), 5.72 (1 H, ddt, $J = 16.9, 10.3$ and 6.7 Hz), 5.22 (1 H, d, $J = 12.4$ Hz), 5.13 (1 H, d, $J = 12.4$ Hz), 5.05 (1 H, d, $J = 8.3$ Hz), 5.02-4.92 (2 H, m), 4.30-4.39 (1 H, m), 1.96-2.12 (2 H, m), 1.75-1.88 (1 H, m), 1.58-1.69 (1 H, m), 1.31-1.52 (11 H, m). $^{13}\text{C NMR}$ (100 MHz, CDCl_3) δ ppm: 172.7, 155.4, 137.9, 135.5, 128.6, 128.4, 128.2, 115.1, 79.8, 66.9, 53.4, 33.1, 32.1, 28.3, 24.4; IR (cm^{-1}), 3368, 1745, 1712, 1634, 1501, 1366, 1253, 1165, 1001, 912; m/z (ES+) found MH^+ : 334.2028, $\text{C}_{19}\text{H}_{28}\text{NO}_4$ requires 334.2018.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)oct-7-enoate (11)⁴¹

General procedure **A** using zinc dust (487.5 mg, 7.5 mmol, 2.5 eq.), iodine (152.3 mg, 0.6 mmol, 0.2 eq.), **5** (1.10 g, 3 mmol, 1 eq.), CuBr.DMS (61.5 mg, 0.3 mmol, 0.1 eq.), and allyl chloride (320 μ l, 3.9 mmol, 1.3 eq.) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)oct-7-enoate **11** (311 mg, 1.15 mmol, 38%) as a colorless oil. Zinc insertion took 40 minutes with sonication at 35 °C. $[\alpha]_D$ -17.6 (c 1.25, MeOH); R_f = 0.52 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 5.77 (1 H, ddt, J = 16.9, 10.1 and 6.8 Hz), 4.9-5.05 (3 H, m), 4.25-4.33 (1 H, m), 3.73 (3 H, s), 1.99-2.1 (2 H, m), 1.69-1.86 (1 H, m), 1.53-1.68 (1 H, m), 1.24-1.51 (4 H, m), 1.44 (9 H, s); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 173.4, 155.3, 138.5, 114.6, 79.8, 53.3, 52.2, 33.4, 32.6, 28.4, 28.3, 24.6; IR (cm⁻¹), 3370, 1745, 1720, 1509, 1168; m/z (ES⁺) found MH⁺: 272.1850, C₁₄H₂₆NO₄ requires 272.1862.

General procedure B: Cross metathesis of unsaturated amino acids

A two-necked round bottomed flask with a magnetic stirrer bar was fitted with a condenser equipped with a three-way tap on top and rubber septum. The flask was flame-dried under vacuum and backfilled with nitrogen three times. The flask was allowed to cool, before the unsaturated amino acid **9-11** and enone in dry degassed CH₂Cl₂ (2 mL) were added via syringe. Grubbs 2nd generation catalyst (5 mol% relative to substrate) in dry CH₂Cl₂ (1 mL) was added by syringe, and the reaction mixture was heated at reflux for 7 h, allowed to cool to room temperature and concentrated. The residue was purified by column chromatography using a gradient of 15-35% EtOAc in petroleum ether.

Methyl (2S,5E)-2-([(tert-butoxy)carbonyl]amino)-7-oxooct-5-enoate (13a)

General procedure **B** using **9** (97 mg, 0.4 mmol, 1 eq.), 3-buten-2-one (100 μ l, 1.2 mmol, 3 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave methyl (2*S*, 5*E*)-2-([(*tert*-butoxy)carbonyl]amino)-7-oxooct-5-enoate **13a** (104 mg, 0.36 mmol, 91%) as an oil. [α]_D +40.0 (c 0.4, CHCl₃); R_f = 0.13 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 6.77 (1 H, dt, *J* = 16.0 and 6.7 Hz), 6.09 (1 H, br.d, *J* = 16.0 Hz), 5.08 (1 H, d, *J* = 8.1 Hz), 4.30-4.42 (1H, m), 3.76 (3 H, s), 2.19–2.39 (2 H, m), 2.25 (3 H, s), 1.92-2.11 (1 H, m), 1.74-1.86 (1 H, m), 1.45 (9 H, s); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 198.3, 172.7, 155.3, 146.1, 131.9, 80.1, 52.8, 52.4, 31.3, 28.3, 28.2, 26.9; IR (cm⁻¹), 3359, 1749, 1715, 1673, 1518, 1450, 1371, 1253, 1166; *m/z* (ES⁺) found MH⁺: 286.1648, C₁₄H₂₄NO₅ requires 286.1654.

Methyl (2*S*, 5*E*)-2-([(*tert*-butoxy)carbonyl]amino)-7-oxonon-5-enoate (13b)

General procedure **B** using **9** (97 mg, 0.4 mmol, 1 eq.), 1-penten-3-one (120 μ l, 1.2 mmol, 3 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave methyl (2*S*, 5*E*)-2-([(*tert*-butoxy)carbonyl]amino)-7-oxonon-5-enoate **13b** (110 mg, 0.37 mmol, 92%) as an oil. [α]_D +36.4 (c 0.55, CHCl₃); R_f = 0.17 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 6.76 (1 H, dt, *J* = 15.8 and 6.8 Hz), 6.09 (1 H, d, *J* = 15.8 Hz), 5.13 (1 H, d, *J* = 7.6 Hz), 4.25-4.36 (1 H, m), 3.72 (3 H, s), 2.53 (2 H, q, *J* = 7.3 Hz), 2.17–2.35 (2 H, m), 1.88-2.06 (1 H, m), 1.71-1.82 (1 H, m), 1.41 (9 H, s), 1.06 (3 H, t, *J* = 7.3 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 200.7, 172.8, 155.3, 144.6, 130.7, 80.0, 52.9, 52.4, 33.4, 31.3, 28.3, 28.2, 8.0; IR (cm⁻¹), 3346, 1747, 1708, 1669, 1512, 1448, 1363, 1167; *m/z* (ES⁺) found MH⁺: 300.1800, C₁₅H₂₆NO₅ requires 300.1811.

Methyl (2*S*,5*E*)-2-([(*tert*-butoxy)carbonyl]amino)-7-oxodec-5-enoate (13c)

General procedure **B** using **9** (97 mg, 0.4 mmol, 1 eq.), 1-hexen-3-one (117 μ l, 1 mmol, 2.5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mmol%) in CH₂Cl₂ (3 mL) gave methyl (2*S*,5*E*)-2-([(tert-butoxy)carbonyl]amino)-7-oxodec-5-enoate **13c** (113 mg, 0.36 mmol, 90%) as an oil. $[\alpha]_D +30.7$ (c 0.88, CHCl₃); R_f = 0.17 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 6.78 (1 H, dt, J = 15.8 and 6.8 Hz), 6.11 (1 H, d, J = 15.8 Hz), 5.08 (1 H, d, J = 7.6 Hz), 4.27-4.39 (1 H, m), 3.75 (3 H, s), 2.5 (2 H, t, J = 7.3 Hz), 2.19–2.37 (2 H, m), 1.91-2.08 (1 H, m), 1.69-1.85 (1 H, m), 1.63 (2 H, sext., J = 7.4 Hz), 1.44 (9 H, s), 0.93 (3 H, t, J = 7.5 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 200.4, 172.8, 155.3, 144.7, 130.9, 80.1, 52.9, 52.4, 42.2, 31.3, 28.3, 28.2, 17.6, 13.8; IR (cm⁻¹), 3359, 1750, 1714, 1675, 1521, 1448, 1366, 1249, 1167; m/z (ES⁺) found MH⁺: 314.1972, C₁₆H₂₈NO₅ requires 314.1967.

Benzyl (2*S*,6*E*)-2-([(tert-butoxy)carbonyl]amino)-8-oxonon-6-enoate (14a)

General procedure **B** using **10** (133.4 mg, 0.4 mmol, 1 eq.), 3-buten-2-one (100 μ l, 1.2 mmol, 3 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave benzyl (2*S*,6*E*)-2-([(tert-butoxy)carbonyl]amino)-8-oxonon-6-enoate **14a** (134 mg, 0.36 mmol, 89%) as an oil. $[\alpha]_D -22.0$ (c 0.91, MeOH); R_f = 0.32 (30% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 7.32-7.42 (5 H, m), 6.70 (1 H, dt, J = 16.0 and 6.9 Hz), 6.04 (1 H, d, J = 16.0 Hz), 5.22 (1 H, d, J = 12.2 Hz), 5.14 (1 H, d, J = 12.2, Hz), 5.05 (1 H, d, J = 8.1 Hz), 4.33-4.41 (1 H, m), 2.13-2.31 (5 H, m), 1.77-1.91 (1 H, m), 1.59-1.72 (1 H, m), 1.37-1.57 (2 H, m), 1.44 (9 H, s); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 198.5, 172.5, 155.4, 147.1, 135.3, 131.6, 128.6, 128.5, 128.3, 79.9, 67.1, 53.2, 32.3, 31.7, 28.3, 26.9, 23.7; IR (cm⁻¹), 3342, 1715, 1694, 1673, 1625, 1499, 1364, 1250, 1157; m/z (ES⁺) found MH⁺: 376.2109, C₂₁H₃₀NO₅ requires 376.2124.

Benzyl (2*S*,6*E*)-2-([(tert-butoxy)carbonyl]amino)-8-oxodec-6-enoate (14b)

General procedure **B** using **10** (134 mg, 0.4 mmol, 1 eq.), 1-penten-3-one (200 μ L, 2 mmol, 5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave benzyl (2*S*,6*E*)-2-([(*tert*-butoxy)carbonyl]amino)-8-oxodec-6-enoate **14b** (141 mg, 0.36 mmol, 90%) as an oil. $[\alpha]_D$ -19.5 (c 0.77, MeOH); R_f = 0.18 in (15% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 7.30-7.39 (5 H, m), 6.72 (1 H, dt, J = 16.0 and 6.8 Hz), 6.05 (1 H, br.d, J = 16.0 Hz), 5.21 (1 H, d, J = 12.2 Hz), 5.13 (1 H, d, J = 12.2 Hz), 5.08 (1 H, d, J = 8.3 Hz), 4.30-4.40 (1 H, m), 2.52 (2 H, q, J = 7.3 Hz), 2.11-2.27 (2 H, m), 1.77-1.89 (1 H, m), 1.58-1.71 (1 H, m), 1.36-1.56 (11 H, m), 1.10 (3 H, t, J = 7.3 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 200.8, 172.4, 155.4, 145.6, 135.3, 130.4, 128.6, 128.4, 128.3, 79.8, 67.1, 53.2, 33.3, 32.2, 31.7, 28.3, 23.7, 8.1; IR (cm⁻¹), 3362, 1741, 1696, 1673, 1629, 1499, 1248, 1159; m/z (ES⁺) found MH⁺: 390.2283, C₂₂H₃₂NO₅ requires 390.2280.

Benzyl (2*S*,6*E*)-2-([(*tert*-butoxy)carbonyl]amino)-8-oxoundec-6-enoate (14c**)**

General procedure **B** using **10** (133.4 mg, 0.4 mmol, 1 eq.), 1-hexen-3-one (117 μ L, 1 mmol, 2.5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ gave benzyl (2*S*,6*E*)-2-([(*tert*-butoxy)carbonyl]amino)-8-oxoundec-6-enoate **14c** (140 mg, 0.35 mmol, 86%) as an oil. $[\alpha]_D$ -30.0 (c 0.1, MeOH); R_f = 0.22 (15% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 7.33-7.41 (5 H, m), 6.73 (1 H, dt, J = 16.0 and 6.8 Hz), 6.06 (1 H, br.d, J = 16.0), 5.22 (1 H, d, J = 12.2 Hz), 5.14 (1 H, d, J = 12.2 Hz), 5.04 (1 H, br.d, J = 8.3 Hz), 4.32-4.42 (1 H, m), 2.48 (2 H, t, J = 7.3 Hz), 2.13-2.27 (2 H, m), 1.78-1.91 (1 H, m), 1.57-1.71 (3 H, m), 1.37-1.56 (11 H, m), 0.93 (3 H, t, J = 7.5 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 200.6, 172.5, 155.5, 145.7, 135.3, 130.7, 128.6, 128.5, 128.3, 80.2, 67.1, 53.2, 42.1, 32.3, 31.7, 28.3, 23.7, 17.6, 13.8; IR (cm⁻¹), 3357, 1747, 1712, 1675, 1629, 1499, 1457, 1365, 1256, 1162; m/z (ES⁺) found MH⁺: 404.2426, C₂₃H₃₄NO₅ requires 404.2437.

Methyl (2*S*,7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxodec-7-enoate (15a)

General procedure **B** using **11** (108.5 mg, 0.4 mmol, 1 eq.), 3-buten-2-one (85 μ l, 1 mmol, 2.5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave methyl (2*S*,7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxodec-7-enoate **15a** (103 mg, 0.33 mmol, 82%) as an oil. $[\alpha]_D^{+20.0}$ (c 0.95, CHCl₃); R_f = 0.25 (30% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 6.76 (1 H, dt, J = 16.0 and 6.9 Hz), 6.05 (1 H, d, J = 16.0 Hz), 5.04 (1H, br.d, J = 8.1 Hz), 4.24-4.34 (1 H, m), 3.73 (3 H, s), 2.17-2.26 (2 H, m), 2.23 (3 H, s), 1.75-1.87 (1 H, m), 1.56-1.68 (1 H, m), 1.29-1.55 (4 H, m), 1.43 (9 H, s); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 198.6, 173.2, 155.3, 147.7, 131.5, 79.9, 53.2, 52.3, 32.5, 32.1, 28.3, 27.6, 26.9, 24.8; IR (cm⁻¹), 3356, 1749, 1717, 1674, 1523, 1441, 1369, 1258, 1164; m/z (ES⁺) found MH⁺: 314.1956, C₁₆H₂₈NO₅ requires 314.1967.

Methyl (2*S*, 7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxoundec-7-enoate (15b)

General procedure **B** using **11** (108.5 mg, 0.4 mmol, 1 eq.), 1-penten-3-one (100 μ l, 1 mmol, 2.5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol %) in CH₂Cl₂ (3 mL) gave methyl (2*S*,7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxoundec-7-enoate **15b** (108 mg, 0.33 mmol, 82%) as an oil. $[\alpha]_D^{+23.7}$ (c 0.93, CHCl₃); R_f = 0.22 (20% EtOAc in petroleum ether); ¹H NMR (500 MHz, CDCl₃) δ ppm: 6.79 (1 H, dt, J = 16.0 and 6.9 Hz), 6.08 (1 H, br.d, J = 16.0 Hz), 5.02 (1 H, br.d, J = 7.5 Hz), 4.25-4.34 (1 H, m), 3.74 (3 H, s), 2.55 (2 H, q, J = 7.3 Hz), 2.21 (2 H, dq, J = 1.3 and 7.2 Hz), 1.73-1.87 (1 H, m), 1.58-1.68 (1 H, m), 1.28-1.55 (4 H, m), 1.44 (9 H, s), 1.09 (3 H, t, J = 7.3 Hz); ¹³C NMR (125 MHz, CDCl₃) δ ppm: 201.0, 173.2, 155.3, 146.2, 130.2, 79.9, 53.2, 52.2, 33.2, 32.5, 32.1, 28.3, 27.6, 24.8, 8.1; IR (cm⁻¹), 3357, 1746, 1715, 1634, 1674, 1514, 1460, 1167; m/z (ES⁺) found MH⁺: 328.2111, C₁₇H₃₀NO₅ requires 328.2124.

Methyl (2*S*,7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxododec-7-enoate (15c)

General procedure **B** using **11** (108.5 mg, 0.4 mmol, 1 eq.), 1-hexen-3-one (117 μ L, 1 mmol, 2.5 eq.) and Grubbs 2nd generation catalyst (16 mg, 0.02 mmol, 5 mol%) in CH₂Cl₂ (3 mL) gave methyl (2*S*,7*E*)-2-(((*tert*-butoxy)carbonyl)amino)-9-oxododec-7-enoate **15c** (125.4 mg, 0.37 mmol, 91%) as an oil. $[\alpha]_D^{+13.2}$ (c 0.38, CHCl₃); R_f = 0.29 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 6.77 (1 H, dt, J = 16.0 and 6.9 Hz), 6.07 (1 H, br.d, J = 16.0 Hz), 5.04 (1 H, br. d, J = 8.3 Hz), 4.24-4.32 (1 H, m), 3.72 (3 H, s), 2.49 (2 H, t, J = 7.3 Hz), 2.19 (2 H, dq, J = 7.2 and 1.3 Hz), 1.74-1.85 (1 H, m), 1.56-1.67 (3 H, m), 1.27-1.54 (4 H, m), 1.42 (9 H, s), 0.92 (3 H, t, J = 7.5 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 200.7, 173.3, 155.3, 146.4, 130.5, 79.9, 53.2, 52.2, 42.1, 32.5, 32.1, 28.3, 27.6, 24.8, 17.7, 13.8; IR (cm⁻¹), 3357, 1750, 1718, 1671, 1516, 1437, 1369, 1247, 1167; m/z (ES+) found MH⁺: 342.2269, C₁₈H₃₂NO₅ requires 342.2280.

1,12-Dibenzyl (2*S*,6*E*/*Z*,11*S*)-2,11-bis(((*tert*-butoxy) carbonyl) amino))dodec-6-enedioate (16)

General procedure **B**, using benzyl (2*S*)-2-(((*tert*-butoxy)carbonyl)amino)hept-6-enoate **10** (110 mg, 0.33 mmol, 1 eq.) as the starting material and Grubbs 2nd generation catalyst (14 mg, 0.016 mmol, 5 mmol%), gave compound **16** (104 mg, 0.163 mmol, 98%) as an oil $[\alpha]_D^{+13}$ (c 1.5, CHCl₃); R_f = 0.3 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 7.29-7.40 (10 H, m), 5.25-5.36 (2 H, m), 5.21 (2 H, d, J = 12.5 Hz), 5.12 (2 H, d, J = 12.5 Hz), 4.99-5.01 (2 H, m), 4.27-4.43 (2 H, m), 1.87-2.08 (4 H, m), 1.72-1.86 (2 H, m), 1.54-1.71 (2 H, m), 1.19-1.52 (4 H, m), 1.44 (18 H, s); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 172.8, 155.4, 135.5, 130.1, 128.6, 128.4, 128.3, 79.8, 66.9, 53.5, 32.5, 31.9, 28.3, 25.0; IR (cm⁻¹), 3370, 1745,

1717, 1501, 1459, 1250, 1163; **m/z (ES+)** found MH^+ : 639.3638, $C_{36}H_{51}N_2O_8$ requires 639.3645.

General Procedure C: Hydrogenation of cross-metathesis product

A two-necked round bottomed flask with magnetic stirrer bar was fitted with a rubber septum and three-way tap, and was flame-dried under vacuum and back-filled with nitrogen three times. The flask was allowed to cool, and palladium on carbon (10% w/w) (amount specified in each experiment) was added to the flask which was evacuated and back-filled with nitrogen three times. Then nitrogen gas line was replaced with balloon of hydrogen gas, and the cross-metathesis product (1 eq.) was added to the flask as a solution in EtOAc (7 mL) via syringe. The flask was evacuated until the reaction mixture began to boil, and then back-filled with hydrogen gas. This procedure was repeated three more times, and the reaction stirred at room temperature for 1 day. To remove the catalyst the mixture was filtered through Celite® and then washed with EtOAc. The filtrate and washings were combined, and the solvent was removed under reduced pressure.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxooctanoate (17a)

General procedure **C** using **13a** (105 mg, 0.368 mmol, 1 eq.) and 10% w/w palladium on carbon (20 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxooctanoate **17a** (100 mg, 0.348 mmol, 94%) as a colorless oil. $[\alpha]_D +15.8$ (c 0.95, $CHCl_3$); $R_f = 0.31$ (40% EtOAc in petroleum ether); 1H NMR (400 MHz, $CDCl_3$) δ ppm: 5.02 (1 H, d, $J = 8.1$ Hz), 4.23-4.34 (1 H, m), 3.74 (3 H, s), 2.43 (2 H, t, $J = 7.2$ Hz), 2.13 (3 H, s), 1.73-1.87 (1 H, m), 1.52-1.68 (3 H, m), 1.44 (9 H, s), 1.22-1.40 (2 H, m); ^{13}C NMR (100 MHz, $CDCl_3$) δ ppm: 208.6, 173.3, 155.4, 79.8, 53.3, 52.3, 43.3, 32.5, 29.8, 28.3, 24.8, 23.2; IR (cm^{-1}), 3370, 1749, 1715, 1516, 1439, 1364, 1250, 1169; **m/z (ES+)** found MNa^+ : 310.1618, $C_{14}H_{25}NO_5Na$ requires 310.1630.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxononanoate (17b)

General procedure **C** using **13b** (58 mg, 0.194 mmol, 1 eq.) and 10% w/w palladium on carbon (10 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxononanoate **17b** (58 mg, 0.193 mmol, 99%) as a colorless oil. $[\alpha]_D^{+16.0}$ (c 1, CHCl₃); R_f = 0.22 (20% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 5.03 (1 H, d, J = 8.1 Hz), 4.21-4.31 (1 H, m), 3.70 (3 H, s), 2.34-2.43 (4 H, m), 1.69-1.84 (1 H, m), 1.50-1.66 (3 H, m), 1.42 (9 H, s), 1.22-1.37 (2 H, m), 1.02 (3 H, t, J = 7.3 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 211.3, 173.3, 155.4, 79.8, 53.3, 52.2, 41.9, 35.9, 32.6, 28.3, 24.9, 23.3, 7.8; IR (cm⁻¹), 3360, 1751, 1712, 1519, 1455, 1370, 1256, 1167; m/z (ES⁺) found: MH⁺ 302.1955, C₁₅H₂₈NO₅ requires 302.1967.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxodecanoate (17c)

General procedure **C** using **13c** (89 mg, 0.284 mmol, 1 eq.) and 10% w/w palladium on carbon (40 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-7-oxodecanoate **17c** (88 mg, 0.279 mmol, 98%) as colorless oil. $[\alpha]_D^{+15.0}$ (c 2, CHCl₃); R_f = 0.46 (30% EtOAc in petroleum ether); ¹H NMR (400 MHz, CDCl₃) δ ppm: 5.01 (1 H, d, J = 8.1 Hz), 4.23-4.34 (1 H, m), 3.73 (3 H, s), 2.39 (2 H, t, J = 7.3 Hz), 2.36 (2 H, t, J = 7.3 Hz), 1.71-1.87 (1 H, m), 1.51-1.67 (5 H, m), 1.44 (9 H, s), 1.24-1.39 (2 H, m), 0.91 (3 H, t, J = 7.3 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 210.8, 173.3, 155.4, 79.8, 53.3, 52.2, 44.7, 42.3, 32.6, 28.3, 24.9, 23.3, 17.3, 13.7; IR (cm⁻¹), 3370, 1749, 1715, 1514, 1454, 1367, 1250, 1170; m/z (ES⁺) found MH⁺: 316.2134, C₁₆H₃₀NO₅ requires 316.2124.

(2S)-2-([(tert-Butoxy)carbonyl]amino)-8-oxononanoic acid (18a)¹⁷

General procedure **C** using **14a** (50 mg, 0.133 mmol, 1 eq.) and 10% w/w palladium on carbon (25 mg) gave (2S)-2-([(tert-butoxy)carbonyl]amino)-8-oxononanoic acid **18a** (34 mg,

0.12 mmol 90%) as a colorless oil. $[\alpha]_D^{+5.0}$ (c 1.0, CHCl_3); $R_f = 0.17$ (EtOAc : petroleum ether : acetic acid, 5 : 5 : 0.1 mL); $^1\text{H NMR}$ (400 MHz, CDCl_3) δ ppm: 5.02 (1 H, d, $J = 7.8$ Hz), 4.25-4.35 (1 H, m), 2.44 (2 H, t, $J = 7.3$ Hz), 2.14 (3 H, s), 1.79-1.93 (1 H, m), 1.63-1.74 (1 H, m), 1.58 (2 H, quint., $J = 7.5$ Hz), 1.27-1.48 (4 H, m), 1.45 (9 H, s); $^{13}\text{C NMR}$ (100 MHz, CDCl_3) δ ppm: 209.6, 176.9, 155.6, 80.1, 53.3, 43.5, 32.3, 29.8, 28.6, 28.3, 25.0, 23.4. IR (cm^{-1}), 3340, 1730, 1700, 1658, 1520, 1390, 1368, 1252, 1167; m/z (ES+) found MH^+ : 288.1806, $\text{C}_{14}\text{H}_{26}\text{NO}_5$ requires 288.1811. In the $^1\text{H NMR}$ the carboxylic acid proton was not observed due, presumably, to its broadness.

(2S)-2-([(tert-Butoxy)carbonyl]amino)-8-oxodecanoic acid (18b)^{13,14,17,18}

General procedure **C** using **14b** (52 mg, 0.134 mmol, 1 eq.) and 10% w/w palladium on carbon (23 mg) gave (2S)-2-([(tert-butoxy)carbonyl]amino)-8-oxodecanoic acid **18b** (35 mg, 0.12 mmol, 89%) as a colorless oil. $[\alpha]_D^{-37.2}$ (c 0.94, CHCl_3); $R_f = 0.28$ (EtOAc : petroleum ether : acetic acid, 5 : 5 : 0.1 mL); $^1\text{H NMR}$ (400 MHz, CDCl_3) δ ppm: 5.02 (1 H, d, $J = 8.0$ Hz), 4.24-4.35 (1 H, m), 2.42 (2 H, q, $J = 7.3$ Hz), 2.41 (2 H, t, $J = 7.3$ Hz), 1.79-1.92 (1 H, m), 1.63-1.74 (1 H, m), 1.58 (2 H, quint., $J = 7.3$ Hz), 1.27-1.48 (4 H, m), 1.45 (9 H, s), 1.05 (3 H, t, $J = 7.3$ Hz); $^{13}\text{C NMR}$ (100 MHz, CDCl_3) δ ppm: 212.1, 176.9, 155.6, 80.2, 53.3, 42.1, 35.9, 32.2, 28.7, 28.3, 25.1, 23.5, 7.8; IR (cm^{-1}), 3322, 1735, 1713, 1681, 1510, 1460, 1395, 1249, 1166; m/z (ES+) found MH^+ : 302.1953, $\text{C}_{15}\text{H}_{28}\text{NO}_5$ requires 302.1967.

(2S)-2-([(tert-Butoxy)carbonyl]amino)-8-oxoundecanoic acid (18c)

General procedure **C** using **14c** (82 mg, 0.2 mmol, 1 eq.) and 10% w/w palladium on carbon (37 mg) gave (2S)-2-([(tert-butoxy)carbonyl]amino)-8-oxoundecanoic acid **18c** (62 mg, 0.196 mmol, 98%) as a colorless oil. $[\alpha]_D^{-16.0}$ (c 0.5, CHCl_3); $R_f = 0.25$ (EtOAc : petroleum ether : acetic acid, 5 : 5 : 0.1 mL); $^1\text{H NMR}$ (400 MHz, CDCl_3) δ ppm: 5.03 (1 H, d, $J = 8.1$ Hz), 4.24-

4.35 (1 H, m), 2.40 (2 H, t, $J = 7.8$ Hz), 2.38 (2 H, t, $J = 7.5$ Hz), 1.77-1.93 (1 H, m), 1.52-1.74 (5 H, m), 1.24-1.51 (4 H, m), 1.45 (9 H, s), 0.91 (3 H, t, $J = 7.4$ Hz); ^{13}C NMR (100 MHz, CDCl_3) δ ppm: 211.6, 176.9, 155.6, 80.2, 53.3, 44.7, 42.5, 32.2, 28.7, 28.3, 25.1, 23.4, 17.3, 13.7; IR (cm^{-1}), 3346, 1717, 1701, 1688, 1511, 1453, 1366, 1243, 1159; m/z (ES^+) found MH^+ : 316.2119, $\text{C}_{16}\text{H}_{30}\text{NO}_5$ requires 316.2124. In the ^1H NMR the carboxylic acid proton was not observed due, presumably, to its broadness.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxodecanoate (19a)

General procedure **C** using **15a** (50 mg, 0.159 mmol, 1 eq.) and 10% w/w palladium on carbon (25 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxodecanoate **19a** (47.5 mg, 0.15 mmol, 94%) as a colorless oil. $[\alpha]_{\text{D}} +10.9$ (c 1.1, CHCl_3); $R_f = 0.17$ (20% EtOAc in petroleum ether); ^1H NMR (500 MHz, CDCl_3) δ ppm: 4.99 (1 H, br.d, $J = 7.7$ Hz), 4.22-4.32 (1 H, m), 3.72 (3 H, s), 2.39 (2 H, t, $J = 7.4$ Hz), 2.12 (3 H, s), 1.69-1.82 (1 H, m), 1.51-1.65 (3 H, m), 1.43 (9 H, s), 1.22-1.37 (6 H, m); ^{13}C NMR (125 MHz, CDCl_3) δ ppm: 209.1, 173.4, 155.3, 79.8, 53.3, 52.1, 43.6, 32.6, 29.8, 28.9, 28.8, 28.3, 25.0, 23.6; IR (cm^{-1}), 3365, 1745, 1712, 1516, 1438, 1370, 1249, 1164; m/z (ES^+) found MH^+ : 316.2137, $\text{C}_{16}\text{H}_{30}\text{NO}_5$ requires 316.2124. In the ^1H NMR the carboxylic acid proton was not observed due, presumably, to its broadness.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxoundecanoate (19b)

General procedure **C** using **15b** (49 mg, 0.15 mmol, 1 eq.) and 10% w/w palladium on carbon (17 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxoundecanoate **19b** (49 mg, 0.148 mmol, 99%) as a colorless oil. $[\alpha]_{\text{D}} +19.3$ (c 0.68, CHCl_3); $R_f = 0.27$ (20% EtOAc in petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ ppm: 5.00 (1 H, br.d, $J = 8.1$ Hz), 4.22-4.31 (1 H, m), 3.72 (3 H, s, OCH_3), 2.34-2.44 (4 H, m), 1.69-1.82 (1 H, m), 1.49-1.65 (3 H, m), 1.43 (9

H, s), 1.20-1.37 (6 H, m), 1.03 (3 H, t, $J = 7.3$ Hz); ^{13}C NMR (100 MHz, CDCl_3) δ ppm: 211.7, 173.4, 155.3, 79.8, 53.3, 52.2, 42.2, 35.8, 32.7, 28.9 (2 C), 28.3, 25.1, 23.7, 7.8; IR (cm^{-1}), 3367, 1746, 1711, 1704, 1518, 1455, 1364, 1168; m/z (ES+) found MH^+ : 330.2290, $\text{C}_{17}\text{H}_{32}\text{NO}_5$ requires 330.2280.

Methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxododecanoate (19c)

General procedure C using **15c** (44 mg, 0.129 mmol, 1 eq.) and 10% w/w palladium on carbon (17 mg) gave methyl (2S)-2-([(tert-butoxy)carbonyl]amino)-9-oxododecanoate **19c** (44 mg, 0.128 mmol, 99%) as a colorless oil. $[\alpha]_D^{+14.1}$ (c 0.85, CHCl_3); $R_f = 0.57$ (30% EtOAc in petroleum ether); ^1H NMR (400 MHz, CDCl_3) δ ppm: 5.00 (1 H, br.d, $J = 8.3$ Hz), 4.23-4.33 (1 H, m), 3.73 (3 H, s), 2.37 (2 H, t, $J = 7.3$ Hz), 2.36 (2 H, t, $J = 7.3$ Hz), 1.69-1.85 (1 H, m), 1.49-1.67 (5 H, m), 1.44 (9 H, s), 1.21-1.37 (6 H, m), 0.90 (3 H, t, $J = 7.5$ Hz); ^{13}C NMR (100 MHz, CDCl_3) δ ppm: 211.4, 173.4, 155.4, 79.8, 53.4, 52.2, 44.7, 42.7, 32.7, 28.9 (2 C), 28.3, 25.1, 23.6, 17.3, 13.7; IR (cm^{-1}), 3370, 1749, 1713, 1692, 1520, 1462, 1247, 1172; m/z (ES+) found MH^+ : 344.2437, $\text{C}_{18}\text{H}_{34}\text{NO}_5$ requires 344.2437.

General Procedure D: Intramolecular aza-Michael reactions

The cross-metathesis product **13** (1 eq.) was dissolved in CH_2Cl_2 (2 mL) and a solution of HCl in Et_2O (1 M, 0.6 mol%) was added. After 19 h stirring the resulting solution was concentrated under reduced pressure to give the product.

1-tert-Butyl 2-methyl (2S,5R)-5-(2-oxopropyl)pyrrolidine-1,2-dicarboxylate (20a) & 1-tert-butyl 2-methyl (2S,5S)-5-(2-oxopropyl)pyrrolidine-1,2-dicarboxylate (20b)

General procedure D using methyl (2S,5E)-2-([(tert-butoxy)carbonyl]amino)-7-oxooct-5-enoate **13a** (32 mg, 0.112 mmol, 1 eq.) and 1 M HCl/ Et_2O (6.6×10^{-4} mmol, 0.66 μL , 0.6

mol%) in CH₂Cl₂ (2 mL), after 19 h stirring at r.t. gave the title compounds **20a/b** (32 mg, 0.11 mmol, 100%), as an oil, in a ratio of (0.04 *cis* **20a**, *t_r* 11.82 min: 0.96 *trans* **20b**, *t_r* 11.67 min) based on GC and ¹H NMR. [α]_D -54.0 (c 1, CHCl₃); R_f = 0.19 (20% EtOAc in petroleum ether); ¹H NMR (500 MHz, DMSO 100 °C) δ ppm: 4.14-4.23 (2 H, m), 3.65 (3 H, s), 2.85 (1 H, br.d, *J* = 16.2 Hz), 2.53 (1 H, dd, *J* = 16.2 and 9.7 Hz), 2.17-2.29 (1 H, m), 2.09 (3 H, s), 1.95-2.06 (1 H, m), 1.78-1.85 (1 H, m), 1.55-1.64 (1 H, m), 1.36 (9 H, s); ¹³C NMR (125 MHz, DMSO) δ ppm: 207.5 (207.4), 173.5 (173.0), 153.1 (153.4), 79.5 (79.8), 59.4 (59.1), 54.1 (53.9), 52.3 (52.2), 47.2 (48.1), 30.6 (30.7), 28.6 (29.3), 28.3 (28.4), 27.9 (27.1); IR (cm⁻¹), 1752, 1703, 1396, 1210, 1165, 1126; *m/z* (ES⁺) found MH⁺: 286.1661, C₁₄H₂₄NO₅ requires 286.1654. In the ¹³C NMR the signals due to the minor rotamer are in parentheses.

1-tert-Butyl 2-methyl (2*S*,5*R*)-5-(2-oxobutyl)pyrrolidine-1,2-dicarboxylate (21a) & 1-tert-butyl 2-methyl (2*S*,5*S*)-5-(2-oxobutyl)pyrrolidine-1,2-dicarboxylate (21b)

General procedure **D** using methyl (2*S*,5*E*)-2-([(tert-butoxy)carbonyl]amino)-7-oxonon-5-enoate **13b** (30 mg, 0.1 mmol, 1 eq.) and 1 M HCl/Et₂O (6 × 10⁻⁴ mmol, 0.6 μL, 0.6 mol%) in CH₂Cl₂ (2 mL), after 19 h stirring at r.t. gave the title compounds **21a/b** (30 mg, 100%) as a solid, in a ratio of (0.02 *cis* , *t_r* 17.69 min: 0.98 *trans* , , *t_r* 17.40 min) based on GC of the crude product, mp 52-54 °C; [α]_D -50.5 (c 0.46, CHCl₃); R_f = 0.22 (20% EtOAc in petroleum ether); ¹H NMR (500 MHz, DMSO 100 °C) δ ppm: 4.22-4.16 (2 H, m), 3.65 (3 H, s), 2.84 (1 H, br.dd, *J* = 16.0 and 3 Hz), 2.52 (1 H, dd, *J* = 16.0 and 9.2 Hz), 2.41 (2 H, q, *J* = 7.4 Hz), 2.19-2.31 (1 H, m), 1.96-2.08 (1 H, m), 1.78-1.85 (1 H, m), 1.55-1.63 (1 H, m), 1.36 (9 H, s), 0.96 (3 H, t, *J* = 7.3); ¹³C NMR (125 MHz, DMSO) δ ppm: 209.9 (209.8), 173.5 (173.0), 153.1 (153.4), 79.5 (79.7), 59.4 (59.1), 54.2 (54.0), 52.3 (52.2), 46.0 (46.7), 35.8 (35.9), 28.6 (29.3), 28.3 (28.4), 27.9

(27.1), 7.94 (7.98); IR (cm⁻¹), 1745, 1705, 1396, 1366, 1212, 1175, 1123; m/z (ES+) found MH⁺: 300.1818, C₁₅H₂₆NO₅ requires 300.1811. In the ¹³C NMR signals due to the minor rotamer are in parentheses.

1-tert-Butyl 2-methyl (2S,5R)-5-(2-oxopentyl)pyrrolidine-1,2-dicarboxylate (22a) & 1-tert-Butyl 2-methyl (2S,5S)-5-(2-oxopentyl)pyrrolidine-1,2-dicarboxylate (22b)

General procedure **D** using methyl (2S,5E)-2-(((tert-butoxy)carbonyl)amino)-7-oxodec-5-enoate **13c** (49 mg, 0.16 mmol, 1 eq.) and 1 M HCl/Et₂O (9 × 10⁻⁴ mmol, 0.9 μL, 0.6 mol%) in CH₂Cl₂ (2 mL), after 19 h stirring at r.t. gave the title compounds **22a/b** (49 mg, 0.16 mmol, 100%), as an oil, in a ratio of (0.02 *cis*, t_r 25.62 min: 0.98 *trans*, t_r 25.14 min) based on GC of the crude product; [α]_D -56.0 (c 1.25, CHCl₃); R_f = 0.18 (15% EtOAc in petroleum ether); ¹H NMR (500 MHz, DMSO 100 °C) δ ppm: 4.15-4.22 (2 H, m), 3.65 (3 H, s), 2.84 (1 H, br.d, J = 16.1 Hz), 2.52 (1 H, dd, J = 16.1 and 9.5 Hz), 2.38 (2 H, t, J = 7.3 Hz), 2.19-2.30 (1 H, m), 1.96-2.07 (1 H, m), 1.78-1.86 (1 H, m), 1.56-1.64 (1 H, m), 1.52 (2 H, sextet, J = 7.3 Hz), 1.36 (9 H, s), 0.87 (3 H, t, J = 7.4 Hz); ¹³C NMR (125 MHz, DMSO) δ ppm: 209.1 (209.0), 173.0 (172.6), 152.6 (152.9), 79.1 (79.3), 58.9 (58.7), 53.7 (53.5), 51.8 (51.7), 45.8 (46.6), 44.2 (44.3), 28.2 (28.8), 27.8 (27.9), 27.5 (26.6), 16.5 (16.6), 13.5 (13.4); IR (cm⁻¹), 1751, 1699, 1392, 1258, 1085, 1020, 795; m/z (ES+) found MH⁺: 314.1982, C₁₆H₂₈NO₅ requires 314.1967. In the ¹³C NMR signals due to the minor rotamer are in parentheses.

General procedure E: Boc deprotection of pyrrolidines 20b, 21b and 22b

The *N*-Boc protected compound was dissolved in CH₂Cl₂ (4 mL). Neat TFA (50 eq.) relative to the substrate was added and the reaction followed by TLC until the starting material has disappeared. The solvent and excess TFA was then removed under reduced pressure to give

the TFA salts. The salts were not sufficiently thermally stable for melting points to be determined.

(2S,5S)-2-(Methoxycarbonyl)-5-(2-oxopropyl)pyrrolidin-1-ium trifluoroacetate (23)

General procedure E using **20b** (88 mg, 0.31 mmol, 1 eq.) and TFA (1.2 mL, 15.5 mmol, 50 eq.) gave (2S, 5S)-2-(methoxycarbonyl)-5-(2-oxopropyl)pyrrolidin-1-ium trifluoroacetate **23** (94 mg, 0.31 mmol, 100%) as a solid, mp decomp.; $[\alpha]_D -10.0$ (c 1, CHCl₃); $R_f = 0.075$ (20% CHCl₃, 30% petroleum ether and 50% acetonitrile), ¹H NMR (400 MHz, CDCl₃) δ ppm: 4.48-4.75 (1 H, m), 3.94-4.08 (1 H, m), 3.86 (3 H, s), 3.18-3.37 (1 H, m), 2.93-3.13 (1 H, m), 2.53-2.67 (1 H, m), 2.19-2.36 (1 H, m), 2.22 (3 H, s), 2.04-2.18 (1 H, m), 1.85-2.00 (1 H, m); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 206.8, 169.7, 58.8, 57.1, 53.7, 44.1, 29.8, 29.5, 27.8; IR (cm⁻¹), 3425, 1751, 1715, 1676, 1445, 1366, 1245, 1185; m/z (ES+) found M⁺: 186.1125, C₉H₁₆NO₃ requires 186.1130. In the ¹H NMR there are additional signals in the low field range due to the two acidic NH protons, and trace residual CF₃CO₂H. The chemical shifts are very variable depending on conditions.

(2S,5S)-2-(Methoxycarbonyl)-5-(2-oxobutyl)pyrrolidin-1-ium trifluoroacetate (24)

General procedure E using **21b** (43 mg, 0.144 mmol, 1 eq.) and TFA (0.55 mL, 7.2 mmol, 50 eq.) gave (2S,5S)-2-(methoxycarbonyl)-5-(2-oxobutyl)pyrrolidin-1-ium trifluoroacetate **24** (45 mg, 0.143 mmol, 99% yield) as a solid, mp decomp.; $[\alpha]_D -6.1$ (c 1.15, CHCl₃); $R_f = 0.125$ (20% CHCl₃, 30% petroleum ether and 50% acetonitrile); ¹H NMR (400 MHz, CDCl₃) δ ppm: 4.55-4.80 (1 H, m), 3.92-4.04 (1 H, m), 3.88 (3 H, s), 3.22-3.38 (1 H, m), 2.93-3.08 (1 H, m), 2.60-2.72 (1 H, m), 2.41-2.59 (2 H, m), 2.21-2.35 (1 H, m), 2.05-2.18 (1 H, m), 1.91-2.05 (1 H, m), 1.08 (3 H, t, $J = 7.1$ Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 209.2, 169.7, 58.7, 56.9, 53.7,

43.1, 35.7, 30.1, 27.9, 7.3; IR (cm⁻¹), 1752, 1715, 1674, 1446, 1246, 1203, 1176; m/z (ES+) found M⁺: 200.1293, C₁₀H₁₈NO₃ requires 200.1287. In the ¹H NMR there are additional signals in the low field range due to the two acidic NH protons, and trace residual CF₃CO₂H. The chemical shifts are very variable depending on conditions.

(2S,5S)-2-(Methoxycarbonyl)-5-(2-oxopentyl)pyrrolidin-1-ium trifluoroacetate (25)

General procedure E using **22b** (98 mg, 0.313 mmol, 1 eq.) and TFA (1.2 mL, 15.6 mmol, 50 eq.) gave (2S, 5S)-2-(methoxycarbonyl)-5-(2-oxopentyl) pyrrolidin-1-ium trifluoroacetate **25** (100 mg, 0.31 mmol, 99% yield) as a solid, mp decomp.; [α]_D -2.0 (c 1, CHCl₃); R_f = 0.15 (20% CHCl₃, 30% petroleum ether and 50% acetonitrile); ¹H NMR (400 MHz, CDCl₃) δ ppm: 4.62 (1 H, t, J = 7.8 Hz), 3.89-4.04 (1 H, m), 3.87 (3 H, s), 3.29 (1 H, dd, J = 18.7 and 9.1 Hz), 2.97 (1 H, dd, J = 18.7 and 3.8 Hz), 2.56-2.67 (1 H, m), 2.35-2.55 (2 H, m), 2.20-2.34 (1 H, m), 2.03-2.16 (1 H, m), 1.87-2.02 (1 H, m), 1.62 (2 H, sextet, J = 7.3 Hz), 0.92 (3 H, t, J = 7.3 Hz); ¹³C NMR (100 MHz, CDCl₃) δ ppm: 208.9, 169.7, 58.8, 57.2, 53.8, 44.4, 43.2, 29.9, 27.8, 16.8, 13.4; IR (cm⁻¹), 1749, 1715, 1674, 1446, 1246, 1203, 1176; m/z (ES+) found M⁺: 214.1439, C₁₁H₂₀NO₃ requires 214.1443. In the ¹H NMR there are additional signals in the low field range due to the two acidic NH protons, and trace residual CF₃CO₂H. The chemical shifts are very variable depending on conditions.

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Supporting Information

Details for the X-ray structure determinations of compounds **21b**, **23**, **24** and **25**, and a combined cif file; ^1H and ^{13}C NMR spectra for **4**, **5**, **9-11**, **13a-c**, **14a-c**, **15a-c**, **16**, **17a-c**, **18a-c**, **19a-c**, **20b**, **21b**, **22b**, and **23-25**; GC Traces for **20a/20b**, **21a/21b**, and **22a/22b**. This information is available free of charge via the Internet at <http://pubs.acs.org>.

References

- (1) Darkin-Rattray, S. J.; Gurnett, A. M.; Myers, R. W.; Dulski, P. M.; Crumley, T. M.; Allocco, J. J.; Cannova, C.; Meinke, P. T.; Colletti, S. L.; Bednarek, M. A.; Singh, S. B.; Goetz, M. A.; Dombrowski, A. W.; Polishook, J. D.; Schmatz, D. M. *Proc. Natl. Acad. Sci. U. S. A.* **1996**, *93*, 13143.
- (2) Singh, S. B.; Zink, D. L.; Polishook, J. D.; Dombrowski, A. W.; Darkin-Rattray, S. J.; Schmatz, D. M.; Goetz, M. A. *Tetrahedron Lett.* **1996**, *37*, 8077.
- (3) Singh, S. B.; Zink, D. L.; Liesch, J. M.; Mosley, R. T.; Dombrowski, A. W.; Bills, G. F.; Darkin-Rattray, S. J.; Schmatz, D. M.; Goetz, M. A. *J. Org. Chem.* **2002**, *67*, 815.
- (4) Han, J. W.; Ahn, S. H.; Park, S. H.; Wang, S. Y.; Bae, G. U.; Seo, D. W.; Kwon, H. K.; Hong, S.; Lee, H. Y.; Lee, Y. W.; Lee, H. W. *Cancer Res* **2000**, *60*, 6068.
- (5) Kim, D. H.; Kim, M.; Kwon, H. J. *Biochem. Mol. Biol.* **2003**, *36*, 110.
- (6) Colletti, S. L.; Li, C.; Fisher, M. H.; Wyvrat, M. J.; Meinke, P. T. *Tetrahedron Lett.* **2000**, *41*, 7825.
- (7) Meinke, P. T.; Colletti, S. L.; Ayer, M. B.; Darkin-Rattray, S. J.; Myers, R. W.; Schmatz, D. M.; Wyvrat, M. J.; Fisher, M. H. *Tetrahedron Lett.* **2000**, *41*, 7831.
- (8) Meinke, P. T.; Colletti, S. L.; Doss, G.; Myers, R. W.; Gurnett, A. M.; Dulski, P. M.; Darkin-Rattray, S. J.; Allocco, J. J.; Galuska, S.; Schmatz, D. M.; Wyvrat, M. J.; Fisher, M. H. *J. Med. Chem.* **2000**, *43*, 4919.
- (9) Murray, P. J.; Kranz, M.; Ladlow, M.; Taylor, S.; Berst, F.; Holmes, A. B.; Keavey, K. N.; Jaxa-Chamiec, A.; Seale, P. W.; Stead, P.; Upton, R. J.; Croft, S. L.; Clegg, W.; Elsegood, M. R. *J. Bioorg. Med. Chem. Lett.* **2001**, *11*, 773.
- (10) Gallo, P.; Latronico, M. V. G.; Grimaldi, S.; Borgia, F.; Todaro, M.; Jones, P.; Gallinari, P.; De Francesco, R.; Ciliberto, G.; Steinkuhler, C.; Esposito, G.; Condorelli, G. *Cardiovasc. Res.* **2008**, *80*, 416.
- (11) Bajgrowicz, J. A.; El, H. A.; Jacquier, R.; Pigiere, C.; Viallefont, P. *Tetrahedron* **1985**, *41*, 1833.
- (12) Mou, L.; Singh, G. *Tetrahedron Lett.* **2001**, *42*, 6603.
- (13) Linares, M. L.; Agejas, F. J.; Alajarin, R.; Vaquero, J. J.; Alvarez-Builla, J. *Synthesis* **2006**, 2069.
- (14) Rodriguez, M.; Bruno, I.; Cini, E.; Marchetti, M.; Taddei, M.; Gomez-Paloma, L. *J. Org. Chem.* **2006**, *71*, 103.
- (15) Kim, S.; Kim, E. Y.; Ko, H.; Jung, Y. H. *Synthesis-Stuttgart* **2003**, 2194.
- (16) Cooper, T. S.; Laurent, P.; Moody, C. J.; Takle, A. K. *Org. Biomol. Chem.* **2004**, *2*, 265.
- (17) Jones, P.; Altamura, S.; De Francesco, R.; Gonzalez Paz, O.; Kinzel, O.; Mesiti, G.; Monteagudo, E.; Pescatore, G.; Rowley, M.; Verdirame, M.; Steinkuhler, C. *J. Med. Chem.* **2008**, *51*, 2350.
- (18) Montero, A.; Beierle, J. M.; Olsen, C. A.; Ghadiri, M. R. *J. Am. Chem. Soc.* **2009**, *131*, 3033.
- (19) Jackson, R. F. W.; Moore, R. J.; Dexter, C. S.; Elliott, J.; Mowbray, C. E. *J. Org. Chem.* **1998**, *63*, 7875.
- (20) Dunn, M. J.; Jackson, R. F. W.; Pietruszka, J.; Turner, D. *J. Org. Chem.* **1995**, *60*, 2210.
- (21) Rodriguez, A.; Miller, D. D.; Jackson, R. F. W. *Org. Biomol. Chem.* **2003**, *1*, 973.

- (22) Jackson, R. F. W.; Fraser, J. L.; Wishart, N.; Porter, B.; Wythes, M. J. *J. Chem. Soc., Perkin Trans. 1* **1998**, 1903.
- (23) Grubbs, R. H.; Chang, S. *Tetrahedron* **1998**, *54*, 4413.
- (24) Grubbs, R. H. *Tetrahedron* **2004**, *60*, 7117.
- (25) Biagini, S. C. G.; Gibson, S. E.; Keen, S. P. *J. Chem. Soc., Perkin Trans. 1* **1998**, 2485.
- (26) Wang, Z. J.; Spiccia, N. D.; Jackson, W. R.; Robinson, A. J. *J. Pept. Sci.* **2013**, *19*, 470.
- (27) Gebauer, J.; Dewi, P.; Blechert, S. *Tetrahedron Lett.* **2005**, *46*, 43.
- (28) Ryan, S. J.; Zhang, Y.; Kennan, A. J. *Org. Lett.* **2005**, *7*, 4765.
- (29) Mukherjee, J. P.; Sil, S.; Pahari, A. K.; Chattopadhyay, S. K. *Synthesis* **2016**, *48*, 1181.
- (30) Jackson, R. F. W.; Perez-Gonzalez, M. *Org. Synth.* **2005**, *81*, 77.
- (31) Koseki, Y.; Yamada, H.; Usuki, T. *Tetrahedron: Asymmetry* **2011**, *22*, 580.
- (32) Jackson, R. F. W.; Moore, R. J.; Dexter, C. S.; Elliot, J.; Mowbray, C. E. *J. Org. Chem.* **1998**, *63*, 7875.
- (33) Lemen, G. S.; Wolfe, J. P. *Org. Lett.* **2010**, *12*, 2322.
- (34) Chatterjee, A. K.; Choi, T.-L.; Sanders, D. P.; Grubbs, R. H. *J. Am. Chem. Soc.* **2003**, *125*, 11360.
- (35) Schmidtman, F. W.; Benedum, T. E.; McGarvey, G. J. *Tetrahedron Lett.* **2005**, *46*, 4677.
- (36) Wabnitz, T. C.; Spencer, J. B. *Org. Lett.* **2003**, *5*, 2141.
- (37) Wabnitz, T. C.; Yu, J. Q.; Spencer, J. B. *Chem. - Eur. J.* **2004**, *10*, 484.
- (38) Zhong, C.; Wang, Y.-K.; Hung, A. W.; Schreiber, S. L.; Young, D. W. *Org. Lett.* **2011**, *13*, 5556.
- (39) Fustero, S.; Jimenez, D.; Sanchez-Rosello, M.; del Pozo, C. *J. Am. Chem. Soc.* **2007**, *129*, 6700.
- (40) Itoh, F.; Yoshioka, Y.; Yukishige, K.; Yoshida, S.; Ootsu, K.; Akimoto, H. *Chem. Pharm. Bull.* **2000**, *48*, 1270.
- (41) Weiss, G. A.; Valenteckovich, R. J.; Collins, E. J.; Garboczi, D. N.; Lane, W. S.; Schreiber, S. L.; Wiley, D. C. *Proc. Natl. Acad. Sci. U. S. A.* **1996**, *93*, 10945.